

Plant Intelligence Lab - Case Study A

Integrated Biological Decision Engine | Real-data retrospective validation

Forecast -> Uncertainty -> Reliability -> Decision Objective -> Experiment Recommendation

Champion forecast: X15 -> Day-21 regeneration

Out-of-fold R2: 0.863

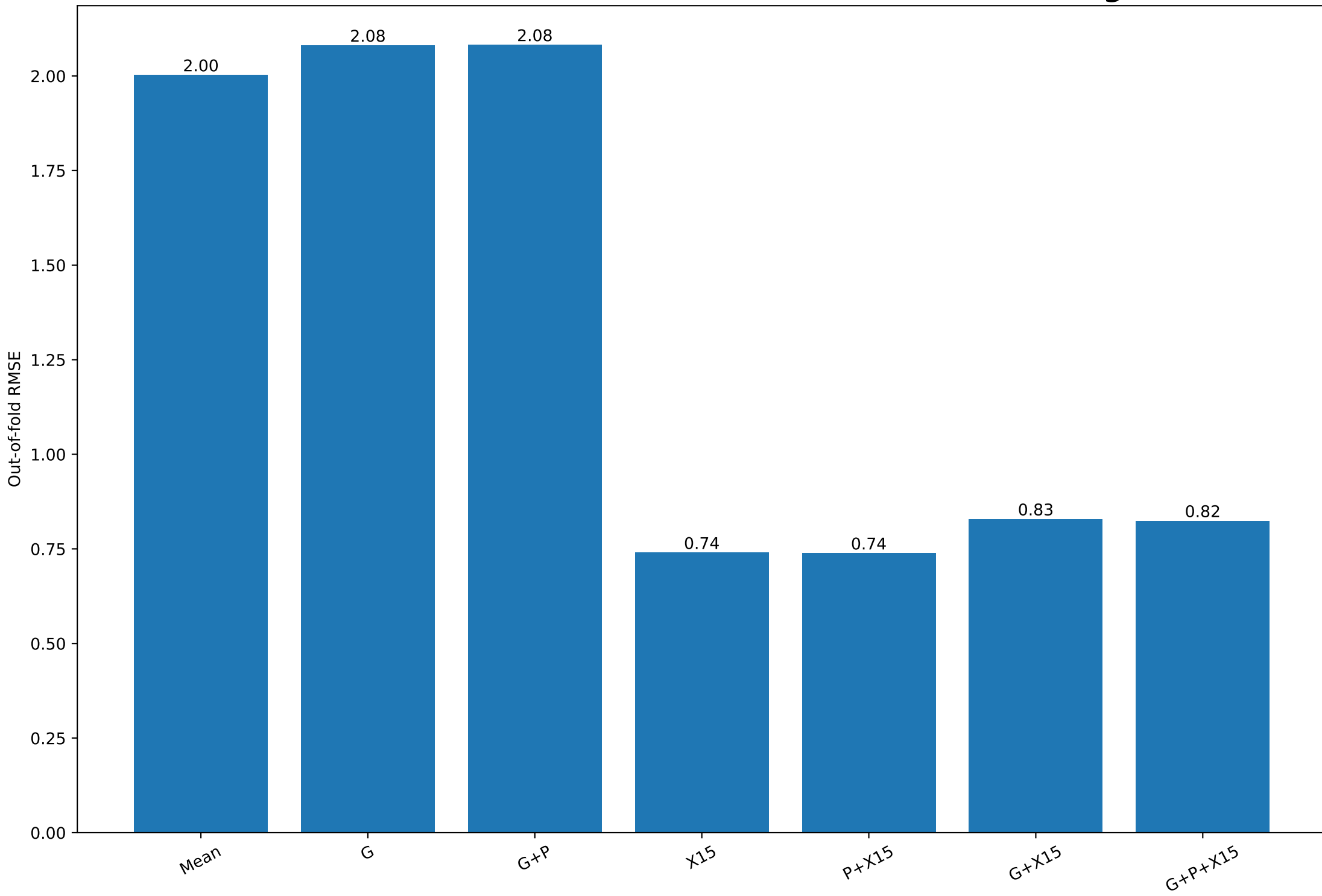
90% empirical coverage: 0.912

Retained RMSE: 0.677

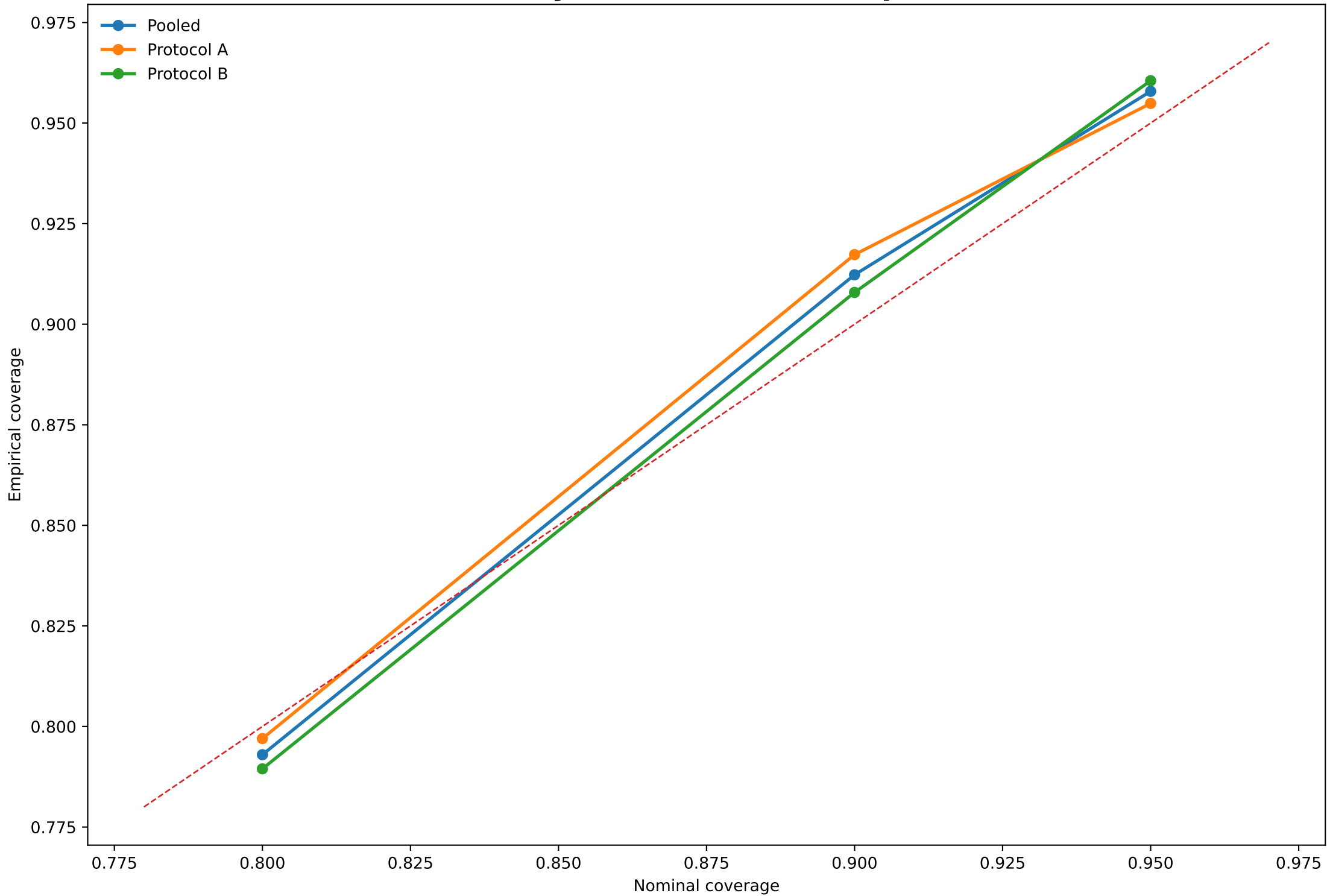
Decision modes: EXPLOIT | EXPLORE | BALANCED

Interpretation: complexity is retained only when it adds measurable decision value.

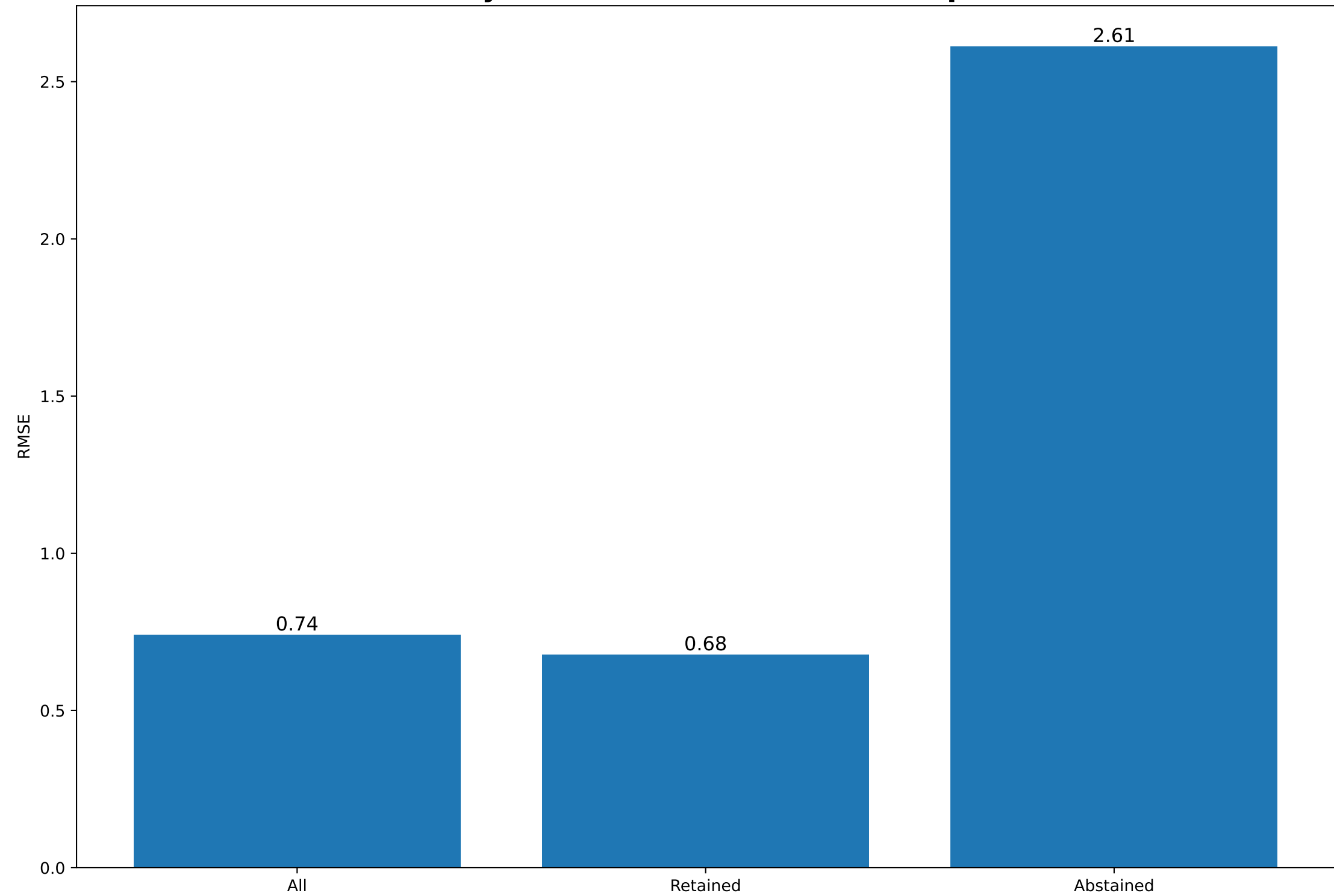
Information ablation: what data are worth collecting?



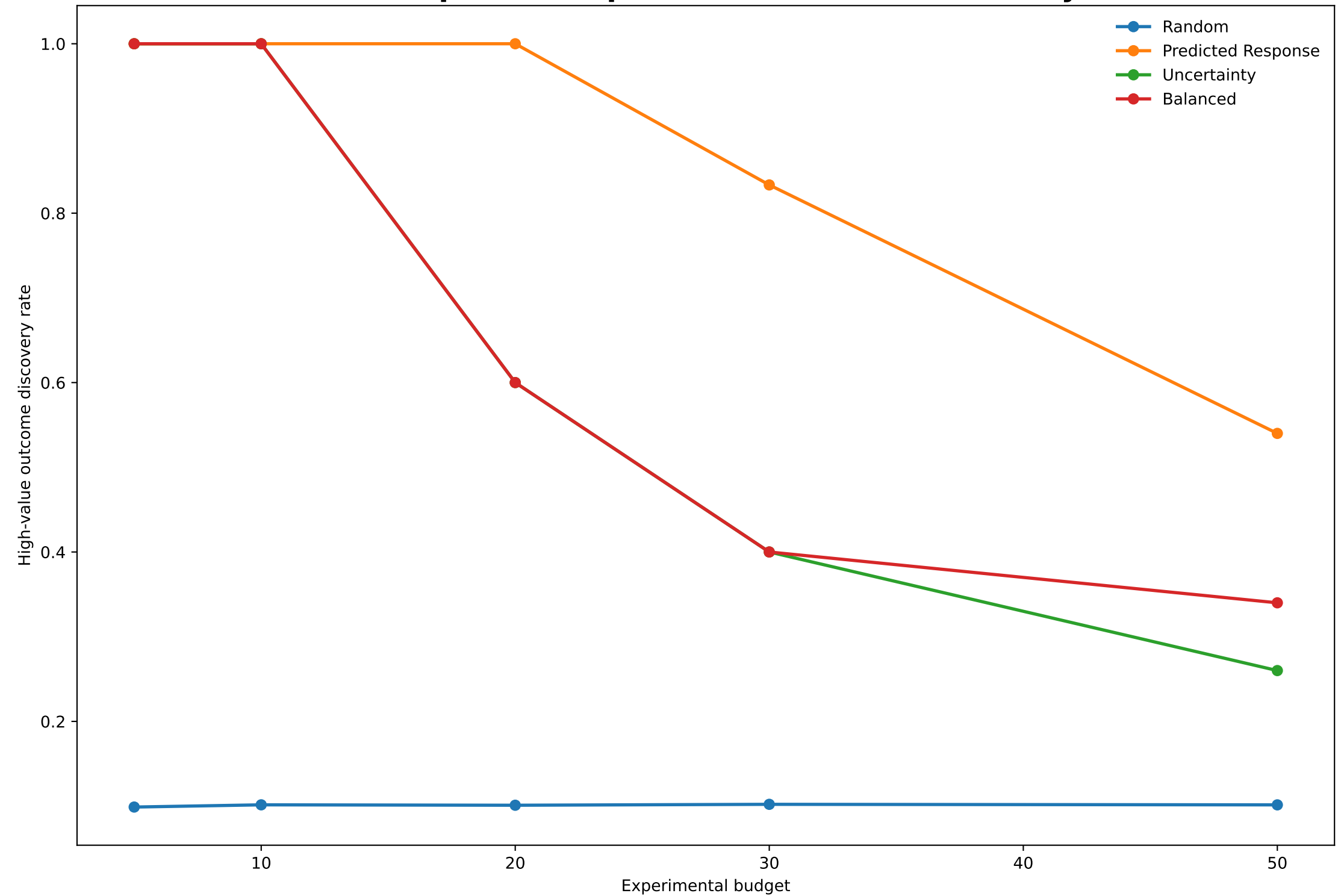
Uncertainty calibration across protocols



Reliability filter: difficult cases are separated



Retrospective experiment-selection efficiency



Decision mode: EXPLOIT

Highest expected Day-21 response among forecasts retained by the reliability filter.

#1 accession 7419 | protocol B | forecast 9.01 | 90% PI [8.10, 9.93] | status FORECAST

#2 accession 7520 | protocol B | forecast 8.41 | 90% PI [7.49, 9.33] | status FORECAST

#3 accession 9764 | protocol B | forecast 8.26 | 90% PI [7.42, 9.11] | status FORECAST

#4 accession 6987 | protocol B | forecast 6.72 | 90% PI [5.80, 7.64] | status FORECAST

#5 accession 7378 | protocol B | forecast 6.20 | 90% PI [5.28, 7.12] | status FORECAST

Retrospective recommendation only; prospective laboratory validation is required before operational claims.

Decision mode: EXPLORE

Highest calibrated predictive uncertainty; prioritizes information rather than immediate response.

#1 accession 7416 | protocol B | forecast 10.93 | 90% PI [10.01, 11.85] | status ABSTAIN

#2 accession 7207 | protocol B | forecast 9.85 | 90% PI [8.94, 10.77] | status ABSTAIN

#3 accession 7419 | protocol B | forecast 9.01 | 90% PI [8.10, 9.93] | status FORECAST

#4 accession 7520 | protocol B | forecast 8.41 | 90% PI [7.49, 9.33] | status FORECAST

#5 accession 6987 | protocol B | forecast 6.72 | 90% PI [5.80, 7.64] | status FORECAST

Retrospective recommendation only; prospective laboratory validation is required before operational claims.

Decision mode: BALANCED

Balances expected response and uncertainty while remaining inside calibration support.

#1 accession 7419 | protocol B | forecast 9.01 | 90% PI [8.10, 9.93] | status FORECAST

#2 accession 7520 | protocol B | forecast 8.41 | 90% PI [7.49, 9.33] | status FORECAST

#3 accession 6987 | protocol B | forecast 6.72 | 90% PI [5.80, 7.64] | status FORECAST

#4 accession 7378 | protocol B | forecast 6.20 | 90% PI [5.28, 7.12] | status FORECAST

#5 accession 7344 | protocol B | forecast 4.99 | 90% PI [4.07, 5.91] | status FORECAST

Retrospective recommendation only; prospective laboratory validation is required before operational claims.